Physiology of the Stomach

MEDICAL PHYSIOLOGY

ILOs

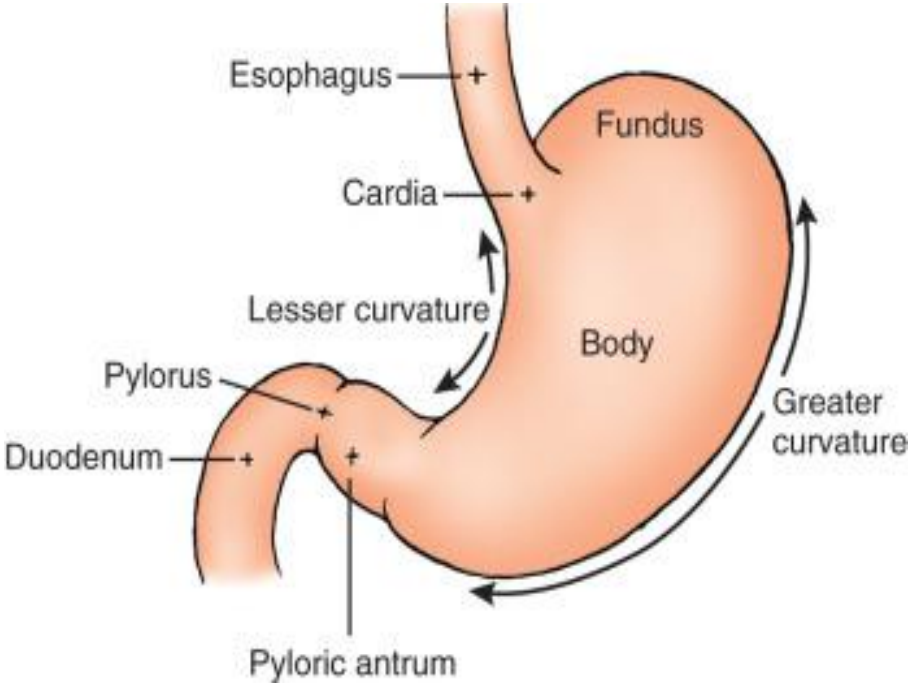
1. Know functions of the stomach.
2. Describe different phases of gastric secretion.
3. Describe gastric secretions and motility.
4. Describe control of gastric functions.
5. Define vomiting and know its causes.
6. Explore mechanism of vomiting and its side effects.

- Food is:
  - Stored in stomach
  - Mixed with acid, mucus, pepsin
  - Released into duodenum

Stomach is divided into:

❶ **Anatomically**

1. Fundus
2. Body
3. Antrum
4. Pylorus



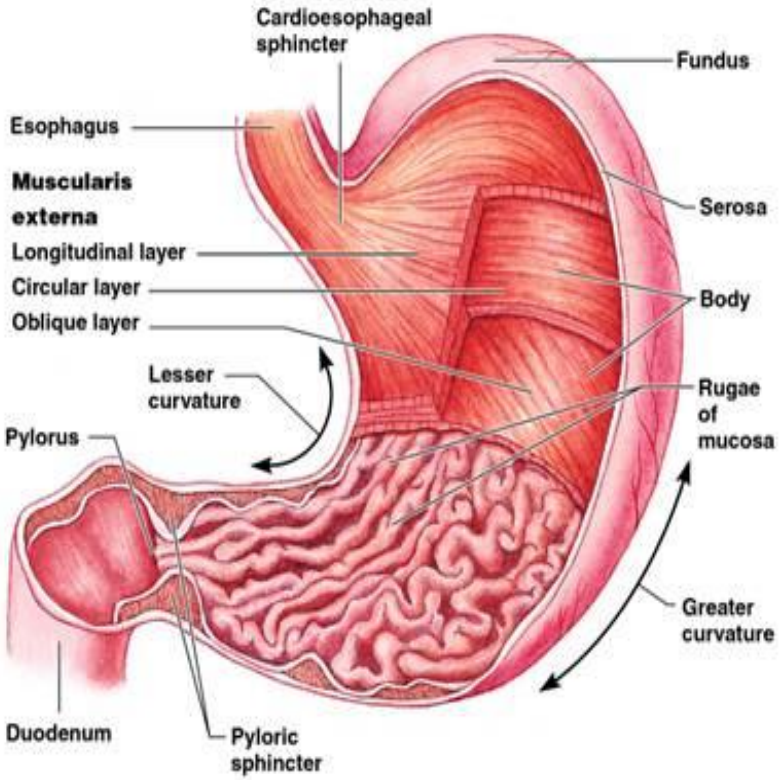
❷ Physiologically

① Proximal motor unit (thin wall)

- Consists of:
  - Fundus
  - Body
- Responsible for: Storage of food (receptive relaxation)

② Distal motor unit (thick wall)

- Consists of:
  - Antrum
  - Pylorus
- Responsible for: Emptying of food (gastric peristalsis)



- <u>Innervation of the stomach:</u>

1-Parasympathetic fibers: Coming from Vagal:

1. Cholinergic **excitatory** fibers (secrete acetyl choline) to distal motor units
2. Purinergic **inhibitory** fibers (secrete ATP) to the proximal motor unit

2-Sympathetic fibers:

1. From lower 6 thoracic segments [via greater splanchnic nerve]
2. They are inhibitory to the proximal motor unit

3-Enteric nervous system formed by the 2 plexus

- **The gastric mucosa contains many deep glands:**

1. In the pyloric & cardiac regions, the glands secrete mucus

2. In the body & fundus, the glands contain:

   a) Parietal cells (oxyntic) **Secrete** HCl & intrinsic factor

   b) Chief cells (peptic=zymogen) **Secrete** pepsinogen

   c) Cells in the neck of the glands **Secrete** mucus which is mixed with HCl & pepsinogen

   d) Mucus cells on the surface of the glands **secrete** mucus & bicarbonate

- **The stomach has a very rich blood & lymphatic supply**

★ Functions of the stomach:

❶ Motor Functions:

1. **Storage function:**
   - It is the function of the **proximal** motor unit which has thin wall
   - Receptive **relaxation**:
     - Occurs in response to swallowing & distension.
     - Stimulus: Distension of the stomach
     - Receptors: Mechanoreceptors
     - Center: 1- Vagal nucleus (in medulla)
       2- **Thoracic** segments of spinal cord
     - Efferent: 1- **Vagal** purinergic fibers (secrete ATP)
       2- **Sympathetic** adrenergic fibers (secrete noradrenalin "NE [Nor Epinephrine]")
       3- **Myenteric plexus** (secrete ATP)
     - Response: Relaxation of stomach (proximal motor unit) with little rise of pressure

2) Mixing & propulsion motility:

Basic electric rhythm (BER) or Gastric slow waves:

- They are spontaneous waves of depolarization.
- They start at midpoint of the greater curvature of the stomach (pacemaker of the stomach) at a rate of 3- 5 cycles/minute. Some leads to spike which is followed by peristaltic waves.

Mechanical response (mixing & emptying):

- Gastric contractions begin in the middle of the body of the stomach and travel towards the pylorus.
- They increase with vagal stimulation or gastrin (from 3 to 5)
- The contractions increase in force & velocity as they travel towards the gastro duodenal junction & and as digestion i.e. by time.

Emptying of stomach into duodenum:

(Regulation of gastric evacuation)

1) Gastric factors
2) Intestinal factors
3) Consistency of food
4) Reflexes from outside the GIT

Emptying of stomach into duodenum: (Regulation of gastric evacuation)

1] Gastric factors: - Gastric distension **increases** stomach emptying

1. Nervous reflexes: ①Short local reflexes
② Long vagovagal reflexes

2. Hormonal: Gastrin

2) Intestinal factors:

A. Nervous (entero gastric reflex):

Through vagus (purinergic fibers): Presence of the following factors in duodenum **inhibits** gastric emptying

1) Mechanical distension
2) Chemical stimuli: eg increased acidity
3) Hyper tonicity
4) Fats
5) Proteins
6) Irritants

B. Hormonal

The presence of fats in the duodenum leads to release of some GIT hormones collectively called **enterogastrone** (**CCK - secretin - GIP**) which **decrease activity** of pyloric pump & slightly increase pyloric contraction

3) Consistency of food:

- Fluid is evacuated more rapidly than solids
- Carbohydrate is evacuated more rapidly than **protein**
- Protein more rapidly than **fat**
- Fat is effective in inhibiting gastric emptying

4) Reflexes from outside the GIT:

a. Pain produces reflex **inhibition** of gastric emptying

b. Emotion can either **increase** or **decrease** gastric motility

❷ Secretory Functions of the stomach:

- Volume: **2.5-3** liters/day
- pH: Highly acidic
- Constituents: ❶ Water
  - ❷ Ions:
    - ① H^+
    - ② Cl^-
    - ③ Na^+
    - ④ K^+
  - ❸ Enzymes:
    - ① Pepsinogen
    - ② Gelatinase
    - ③ Lipase
  - ❹ Mucus
  - ❺ Intrinsic factor "IF" for absorption of vitamin B_{12}

Functions of HCl:

❶ Activation of pepsinogen to pepsin

❷ Provide optimum pH for action of **pepsin**

❸ Helps in **absorption of**:
- Iron
- Calcium

❹ Aids **protein** digestion

❺ **Precipitate milk in** stomach
(to be exposed for long time to pepsin)

❻ Dissolves food particles changing to form **chyme**

❼ Maintains relative sterility of the stomach
(ie kills the bacteria)

★ Mechanism of acid secretion:

❶ H_2CO_3 is: Formed by hydration of CO_2 by: Carbonic anhydrase in the parietal cells

❷ H_2CO_3 is: Spitted into H^+ & HCO_3^{--}

$CO_2 + H_2O \xrightarrow[\text{Enzyme}]{\text{Carbonic anhydrase}} H_2CO_3$

Inside the parietal (oxyntic) cell

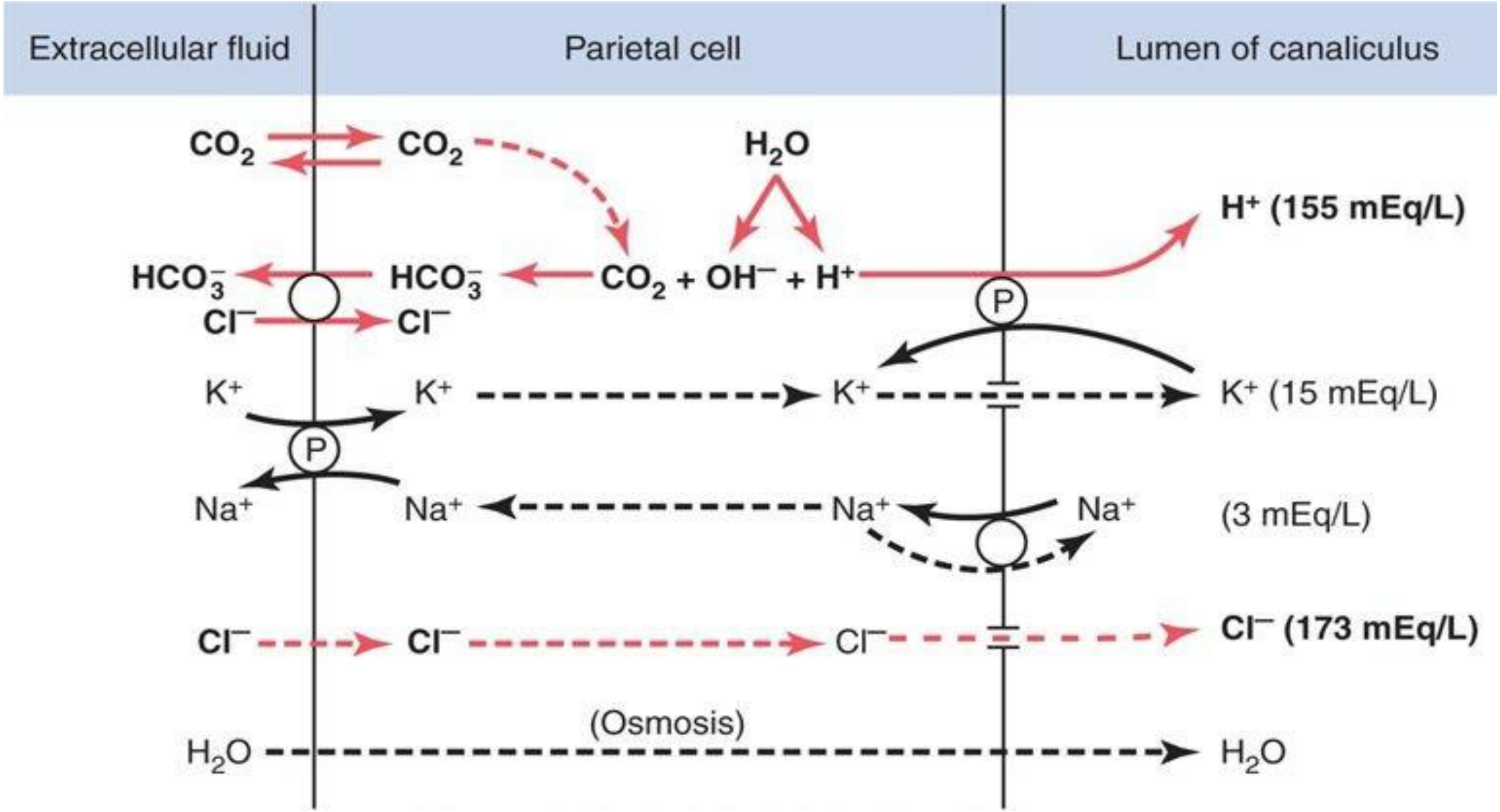
H^+

Pumped out of the cell into gastric lumen [Against concentration gradient (so it needs ATP)]

NB: This occurs in exchange of K^+ through H^+-K^+ ATPase in **apical (luminal) border** of parietal cells

+ HCO_3^{--}

- Is secreted out of the cell (into blood) in exchange with Cl^- (**antiports**) [in **basal border** of parietal cells]
- Cl^- diffuses into gastric lumen down it's electro-chemical gradient through channels
- This is called **Cl^- shift phenomenon**


Extracellular fluid
Parietal cell
Lumen of canaliculus
CO_2
CO_2
H_2O
H^+ (155 mEq/L)
HCO_3^-
HCO_3^-
$CO_2 + OH^- + H^+$
Cl^-
Cl^-
P
K^+
K^+
K^+
K^+ (15 mEq/L)
P
Na^+
Na^+
Na^+
Na^+
(3 mEq/L)
Cl^-
Cl^-
Cl^-
Cl^- (173 mEq/L)
(Osmosis)
H_2O
H_2O

★ **Factors affect acid secretion:**

1-Nervous: **Vagus** & intrinsic Plexus:
- 50% of nerve signals to stomach are through **vagus**
- The other 50% are through local reflexes (**enteric nervous system**)

2-Gastrin:
- Gastrin stimulates HCl secretion
- It is stimulated by:
  - Signals from vagus
  - Local enteric reflexes

3-Histamine:
- Comes from cells in mucosa that resembles mast cells
- Small amount of histamine is a necessary cofactor for exciting significant acid secretion by **acetyl choline** or **gastrin**
- The histamine receptors are:
  - H_2
  - Its mechanism of action is **cAMP**
  - Can be blocked by H_2 blockers as: Cimetidine or Ranitidin

4-Prostaglandin (PGE2): Inhibit acid secretion

Phases of gastric secretion: 3 Phases

❶ **Cephalic Stimulatory Phase:** *Nervous*

❷ **Gastric Stimulatory Phase:** *Nervous & Hormonal*

❸ **Intestinal Inhibitory Phase:** *Nervous & Hormonal*

❶ Cephalic Stimulatory Phase: *Nervous*

- It accounts for ⅓ of the gastric secretion
- Both: - Conditioned
  - Unconditioned reflexes

Stimulate the dorsal motor nucleus of **vagus** →increasing the secretion of:

① HCl

② Pepsinogen

③ Gastrin

④ Mucus

❷ Gastric Stimulatory Phase: *Nervous & Hormonal*

It accounts for ⅔ (70%) gastric secretion

① Nervous: -Long **vagovagal** reflexes

-Short local enteric reflexes

② Hormonal: **Gastrin** hormone

❸ Intestinal Inhibitory Phase: *Nervous & Hormonal*

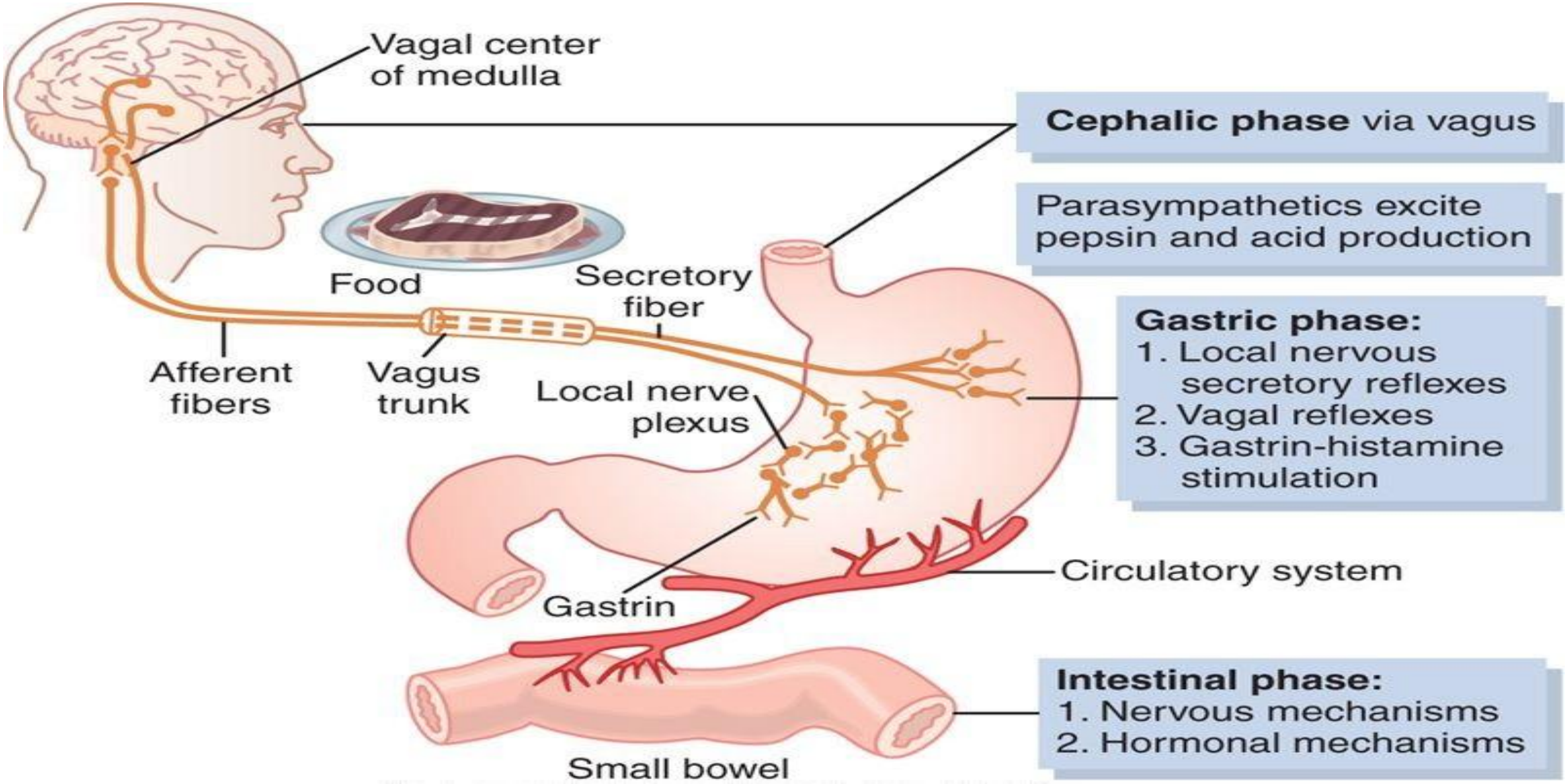
- Fat
- Carbohydrate
- Amino acids

In duodenum **inhibit gastric secretion** (acid & pepsin) via:

① Nervous (Neural): Enterogastric reflex:

② Hormonal (Hormonal mechanism)

- Enterogastrone hormones (secretin, CCK, GIP & VIP)
- NB: **Intestinal gastrin** stimulates gastric secretion


Vagal center of medulla
Cephalic phase via vagus
Parasympathetics excite pepsin and acid production
Food
Secretory fiber
Afferent fibers
Vagus trunk
Local nerve plexus
Gastric phase:
1. Local nervous secretory reflexes
2. Vagal reflexes
3. Gastrin-histamine stimulation
Circulatory system
Gastrin
Intestinal phase:
1. Nervous mechanisms
2. Hormonal mechanisms
Small bowel

Hunger contractions (Hunger pain):

- They are painful gastric contractions between meals, associated with hunger, fasting, they are sometimes felt

- Onset: 12-14 hr from last meal
  Max: 3 days
- The feeding center: In the **hypothalamus** is **inhibited by** impulses from the satiety center
- Hypoglycemia decreases the activity of the satiety center leading to increased activity of the feeding center which sends impulses to:
  1- Limbic cortex leading to **hunger sensation**
  2- Vagal nucleus in the medulla stimulating vagus leading to **hunger contractions**

Vomiting

- **Definition:** It is reflex emptying of gastric content through the mouth
- **Center:** This center is: Medulla which is anatomically & functionally related to the respiratory centers
- **Mechanism:** - Before vomiting:
  (1) Nausea
  (2) Salivation
  (3) Sweating
  (4) Tachycardia

- During vomiting:

1- Stomach:
- It is completely passive
- Relaxation of the wall of the stomach
- Relaxation of the Lower Esophageal Sphincter (LES)
- **Contraction of pyloric antrum**

2- Strong contraction of diaphragm & abdominal muscles:
Leading to increase intra-abdominal pressure which squeezes the content out of stomach

★ **<u>Causes or types:</u>** ❶ Reflex causes:

Afferent in sympathetic & vagus from mucosa of upper GIT

1-Mechanical stimulation of posterior part of the tongue

2- -Gastritis (Irritation of gastric mucosa)

-Peptic Ulcer Disease "PUD"

3-Intestinal obstruction

4-Visceral pain: eg renal colic & appendicitis

❷ Central causes

Stimulate Chemoreceptor Trigger Zone (CTZ) in medulla

1-Drugs: eg: -Apo morphine

-**Emetine**

-**Tarteremetic**

2-Hypoxia

3-Acidosis

4-Uremia

5-Motion sickness: Due to afferents from vestibular nuclei

6-Conditioned reflexes: eg: -Nauseating **smells**

-Sickeing **sights**

- **Effects of vomiting:**

1. Alkalosis: Due to loss of HCl (may lead to tetany)
2. Dehydration:
   ① Hypotension
   ② Tachycardia
3. Hypokalemia (↓K): Due to potassium loss

Thank You